Chemopreventive properties and molecular mechanisms of the bioactive compounds in Hibiscus sabdariffa Linne.

Hibiscus sabdariffa Linne is a traditional Chinese rose tea and has been effectively used in folk medicines for treatment of hypertension, inflammatory conditions. H. sabdariffa aqueous extracts (HSE) were prepared from the dried flowers of H. sabdariffa L., which are rich in phenolic acids, flavonoids and anthocyanins. In this review, we discuss the chemopreventive properties and possible mechanisms of various H. sabdariffa extracts. It has been demonstrated that HSE, H. sabdariffa polyphenol-rich extracts (HPE), H. sabdariffa anthocyanins (HAs), and H. sabdariffa protocatechuic acid (PCA) exert many biologic effects. PCA and HAs protected against oxidative damage induced by tert-butyl droperoxide (t-BHP) in rat primary hepatocytes. In rabbits fed cholesterol and human experimental studies, these studies imply HSE could be pursued as atherosclerosis chemopreventive agents as they inhibit LDL oxidation, foam cell formation, as well as smooth muscle cell migration and proliferation. The extracts also offer hepatoprotection by influencing the levels of lipid peroxidation products and liver marker enzymes in experimental hyperammonemia. PCA has also been shown to inhibit the carcinogenic action of various chemicals in different tissues of the rat. HAs and HPE were demonstrated to cause cancer cell apoptosis, especially in leukemia and gastric cancer. More recent studies investigated the protective effect of HSE and HPE in streptozotocin induced diabetic nephropathy. From all these studies, it is clear that various H. sabdariffa extracts exhibit activities against atherosclerosis, liver disease, cancer, diabetes and other metabolic syndromes. These results indicate that naturally occurring agents such as the bioactive compounds in H. sabdariffa could be developed as potent chemopreventive agents and natural healthy foods.